

37. If $v_1, \dots, v_4$ are in $\mathbb{R}^4$ and $\{v_1, v_2, v_3\}$ is linearly dependent, then $\{v_1, v_2, v_3, v_4\}$ is also linearly dependent.

Solution: From Theorem 7, there exists some vector among v_1, v_2 or v_3 which is a linear combination of the other two vectors. Let, wlog that vector be v_3 .
 $\therefore$ By defn, $\exists c_1, c_2 \in \mathbb{R}$ s.t. $v_3 = c_1 v_1 + c_2 v_2$ $\Rightarrow v_3 = (c_1 v_1 + c_2 v_2) + 0$ (Algebraic property iii)
 $\Rightarrow v_3 = (c_1 v_1 + c_2 v_2) + 0 \cdot v_4$. $\therefore$ By defn, v_3 is a linear combination of v_1, v_2 and v_4 .
 By Theorem 7, $\{v_1, v_2, v_3, v_4\}$ is linearly dependent. $\square$
 $\therefore$ The given statement is True

38. If $v_1, \dots, v_4$ are linearly independent vectors in $\mathbb{R}^4$, then $\{v_1, v_2, v_3\}$ is also linearly independent.

Solution: By defn, $c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 = 0$ if and only if $c_1 = c_2 = c_3 = c_4 = 0$.
 $\therefore c_1 v_1 + c_2 v_2 + c_3 v_3 + 0 \cdot v_4 = 0 \Rightarrow (c_1 v_1 + c_2 v_2 + c_3 v_3) + 0 = 0 \Rightarrow$
 $(c_1 v_1 + c_2 v_2 + c_3 v_3) = 0$ (By Algebraic property iii)

Let, $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ for some $c_1, c_2, c_3 \in \mathbb{R}$.
 $(c_1 v_1 + c_2 v_2 + c_3 v_3) + 0 = 0 + 0$ (Algebraic property (iii))

$(c_1 v_1 + c_2 v_2 + c_3 v_3) + 0 \cdot v_4 = 0$. From defn, $c_1 = c_2 = c_3 = 0$.

The reverse situation is very easy to prove. By defn, $\{v_1, v_2, v_3\}$ is linearly independent. $\therefore$ The given statement is True

39. Suppose A is an $m \times n$ matrix with the property that for all b in $\mathbb{R}^m$ the equation $Ax = b$ has at most one solution. Use the definition of linear independence to explain why the columns of A must be linearly independent.

Solution: Let 0 be the 0 vector in $\mathbb{R}^m$. According to given condition, the matrix eqn. $Ax = 0$ only has the unique trivial solution. ~~the column~~
 Let $A = [a_1 \ a_2 \ \dots \ a_n]$. By Row vector Rule, $x_1 a_1 + x_2 a_2 + \dots + x_n a_n = 0$.
 We know that this is true only if $x_1 = x_2 = \dots = x_n = 0$. By defn, the columns of A must be linearly independent.